



HABIB UNIVERSITY

Data Structures & Algorithms

CS/CE 102/171 Spring 2025

Instructor: Maria Samad

Date: April 21st, 2025

Quiz # 5

Time Allowed: 25 minutes

Max Points: 25

Name: _____

QUESTION: Binary Search Tree

1. For the given set of nodes, construct a binary search tree:

- Nodes = {50, 30, 70, 10, 60, 80, 5, 20, 65, 75, 85, 1, 15, 25, 68, 72, 90}

2. Search for Node 69 in the BST constructed in part (1), by specifying its complete path including its directions:

Ans. _____

3. Search for the maximum in the BST constructed in part (1), by specifying its complete path including its directions:

Ans. _____

4. Find the successor of Node 70 in the BST constructed in part (1), by specifying its complete path including its directions:

Ans. _____

5. Find the predecessor of Node 72 in the BST constructed in part (1), by specifying its complete path including its directions:

Ans. _____

6. Delete the following nodes one after the other from the BST constructed in part (1). Redraw the tree to get complete grade for deletion steps. Use inorder predecessor wherever needed for deletion.

- Node 25:

b. Node 30:

c. Node 50:

7. For the tree given below, write down its Linked Structure Representation:

